

The fan votes of *Dancing with the Stars* are secret, yet recoverable: each week's elimination alone fixes the set of fan vote shares that could explain it, and we report an estimate with how sharply it is pinned down.

Recoverability and certainty. A feasible distribution exists in all 197 elimination weeks and all 25 percentage-method seasons. Certainty is not uniform: a single week leaves shares nearly unconstrained (median interval width 0.992), while smoothness across a season cuts the season-wide median width to 0.16, yet the 90th percentile stays at 0.93 — about one contestant in ten remains unidentifiable. Certainty varies by contestant and week.

Consistency. Replaying our estimates reproduces the observed elimination in 39% of held-out weeks (67 of 172), against a 10% random baseline. A single point in the same model reproduces every elimination of a season (upper bound 1.000), so the shortfall is in the deliberately simple midpoint estimate, not the criterion: 0.390 is that estimator's success rate, not the model's accuracy.

The two combine rules. Applied to every season, the two rules disagree on just 39 of 198 weeks (19.7%). Survivor-weeks correlate with fan shares almost identically under both (0.917 percent, 0.928 rank; rank higher in 6 of 24 seasons), so neither systematically favors the fans. The named controversies in scope are unchanged in direction (season 4: two differing weeks; season 11: none).

Celebrity and dancer effects. Age lowers both the judge share and the fan share (coefficients -0.0012 and -0.0015 per year, each several standard errors from zero); being US-born is same-signed but insignificant. Season and dancer fixed effects leave the signs unchanged: these characteristics act on judges and fans in the same direction (association only).

A better rule. We recommend judge weight about 0.25 plus the bottom-two judge save introduced in season 28: 0.9746 on fairness versus the current 0.9619, and 0.9175 on audience engagement versus 0.9170. The dominance holds only when judges keep the higher-scoring couple; the direction (a save raises fairness) is robust to that tie-break, but under judges keeping the higher fan vote the rule scores 0.9471, below the current 0.9619.

Scope and limitations. All computation covers only the percentage-method seasons (3-27, twenty-five seasons); the rank-method seasons (1-2 and 28 onward) are not covered, so the named season 2 lies outside the scope (season 27 is inside it; its controversy concerns the final placing, not weekly elimination). Fairness and engagement are our own rank-correlation measures; we claim superiority only under them, and only when the bottom-two step is decided on judges' scores. The season-to-rule boundary is uncertain; shifting it one or two seasons moves each reading by about 0.01. A full report and a producer memo accompany this.

Key words: inverse problem; constrained latent variable; linear-programming feasible set; rank correlation; voting-rule design.

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1 Introduction

1.1 Background

Dancing with the Stars (hereafter DWTS) is an American dance competition program in which celebrities are paired with professional dancers. The program has completed 34 seasons. Each week every surviving couple performs, is scored by a panel of expert judges, and is voted on simultaneously by the audience — by telephone or online, for the couples the viewer wishes to keep, with the maximum number of votes announced each week. The judge scores and the audience votes are then combined by the rule the program used that season, and the couple with the lowest combined score is eliminated; in the final week the combined score determines the ranking.

The design question that makes the program interesting is not the dancing but the *combination rule*. The statement records that the rule has changed over time: seasons 1–2 combined the two inputs by rank, seasons 3–27 combined them by percentage, and the program later reverted to ranks and added a further step in which the judges vote to decide the elimination between the two couples at the bottom of the combined result. Each change followed a public controversy in which a contestant with persistently low judge scores survived for a long time — and, in one case, won. Judge scores and audience votes measure different things, a technically strong dance need not be popular and a popular contestant need not be technically strong, so any rule that merges them must trade one off against the other. The statement itself notes that the producers may even welcome a degree of disagreement, since it drives interest.

The audience vote is the quantity at the centre of that trade-off, and it is the one quantity the data does not contain. The statement supplies a dataset covering seasons 1–34 — the identity of each celebrity contestant (occupation, hometown, age at the time of competing), the professional partner, the season’s final result, and the weekly per-judge scores — and states plainly that the audience vote counts are confidential. The scores are recorded from 1 to 10 with decimals (the average when a couple performs more than one dance in a week), already include the allocation of bonus points and team-dance scores; a missing entry is recorded as N/A (either no fourth judge that week, or the week was not aired that season), and a couple’s entries for the weeks after its elimination are recorded as 0. The statement gives one worked example of each combination rule, and states that the exact season in which the program reverted to ranks cannot be verified and can only be reasonably assumed.

1.2 Restatement of the Problem

The problem requires the following five tasks.

1. Build a mathematical model to estimate, for each season, the audience votes of each competing couple in each of the weeks it competed, and give a measure of consistency with the elimination results; also give a measure of certainty for this estimate, and state whether that certainty varies by couple and by week.

2. Use the resulting estimates to compare the two combination rules: apply the rank method and the percentage method to each season separately, compare whether the two produce different results and whether one of the methods better favours the audience vote; for the several controversial cases named in the statement, test whether changing the combination rule would alter the final result, assess the effect of adding the step in which “the bottom two are decided by a judges’ vote,” and finally give a recommendation on which rule should be adopted.
3. Build a model to analyse the influence of professional dancers’ and contestants’ characteristics (age, industry, and so on) on their performance, and answer whether these factors act in the same direction on judge scores and on audience votes.
4. Propose a new combination rule and argue that it is better than the existing rules (fairer, or better by some other criterion), giving the reasons the program should adopt it.
5. Write a complete report of no more than 25 pages, accompanied by a one- to two-page memo addressed to the program’s producers, summarising the conclusions and giving a recommendation on the combination rule.

1.3 Our Work

The estimate at the heart of the problem is an *inverse* one: the audience vote is never observed, but every week it leaves one observable trace, namely which couple went home. We therefore do not treat the vote as a quantity to be predicted, but as a quantity to be *constrained*: for each season, and for every week with an elimination, the statement’s own rule supplies one inequality that any admissible vote distribution must satisfy. Solving those inequalities together yields, per contestant, an interval rather than a number — which is exactly what the second half of task (a) asks for, since the width of that interval is the certainty of the estimate.

Figure 1 shows the pipeline end to end. Section 2 states the variables and the assumptions that make the constraints meaningful; Section 4 writes the model; Section 5 solves it and reports the four quantitative answers (feasibility and certainty, the comparison of the two rules, the effects of contestants’ characteristics, and the proposed mechanism); Section 6 tests how far those answers depend on the choices we had to make, and Section 8 concludes. Two features of the pipeline deserve to be flagged at the outset, because they bound every number that follows. First, the inversion works only where the combination rule is a *linear* function of the vote shares — that is, only for the percentage-method seasons (3–27); the rank-method seasons are drawn with a dashed border in Figure 1 and are not covered. Second, the estimate we obtain is a within-season latent quantity, not any single week’s vote count.

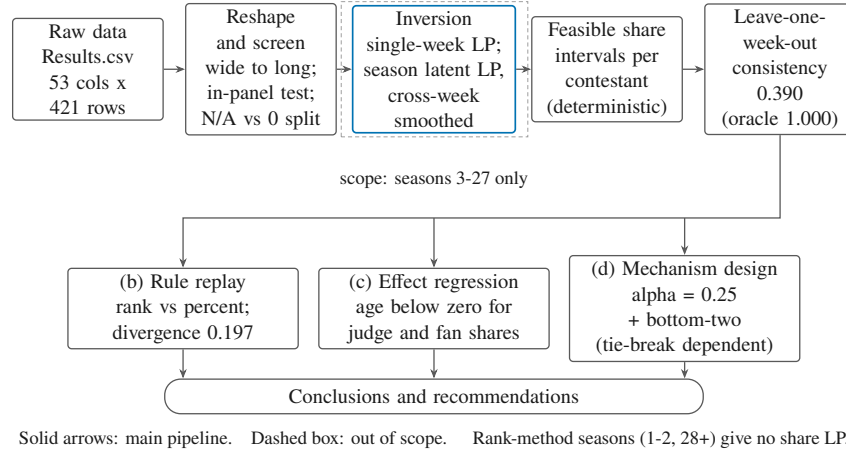


Figure 1: The pipeline, end to end. The attached dataset is reshaped, the weekly elimination constraints are assembled from the season’s combination rule, and linear programming turns them into per-contestant feasible intervals. The intervals feed both the certainty measure and the three downstream questions. The dashed block is the rank-method seasons, which the inversion does not cover.

2 Assumptions and Their Justification

2.1 Variables and Notation

The table below lists the notation used throughout. Every symbol appears here at its first use and is not redefined in later sections.

Table 1: Notation

Symbol	Meaning	Unit
s	season index, $s = 1, \dots, 34$	—
w	week index within the season	—
i	competing couple (celebrity–professional dancer pairing)	—
j	judge index, $j = 1, \dots, 4$	—
$J_{i,s,w}$	total judge score of couple i in week w of season s	points (precision 0.5)
$v_{i,s,w}$	audience vote share of couple i in week w of season s	— ($0 \leq v \leq 1$, summing to 1 over those present that week)
$A_{s,w}$	set of couples still present in week w of season s	—

2.2 Assumptions

Assumption 1 (elimination criterion) Under the combination rule adopted in that season, the couple eliminated that week is the one with the lowest combined score that week; from the season in which the bottom-two step was added, the eliminated couple is one of the two with the lowest combined scores, selected by the judges’ vote between those two.

Assumption 2 (normalization of vote shares and across-week continuity) Within a single week, the audience vote shares of the couples present sum to 1; and the latent popularity of a given couple changes continuously between adjacent weeks, with no unjustified jumps.

Assumption 3 (comparability of judge scores) Within a season, judge scores are on a consistent basis across judges and across weeks, and may be added and compared directly.

Assumption 4 (data semantics) In the dataset, N/A means the cell is not applicable (either no fourth judge that week or the week was not aired that season), while 0 means the couple was eliminated earlier; the two have different meanings and must not be treated together.

Assumption 5 (season mapping of the combination rules) Seasons 1–2 combine by rank, seasons 3–27 combine by percent, and from season 28 onward it reverts to combining by rank together with the added bottom-two step.

2.3 Justification

Assumption 1 This assumption is the observational pivot of the whole paper’s estimation: the audience votes are unobservable, the only usable constraint comes from the elimination results, and it is the season’s combination rule that links the elimination results to the unknown vote counts. An appendix to the statement uses season 1, week 4 and season 5, week 9 as examples to show that the eliminated couple is the one with the lowest combined score, so the assumption is consistent with the statement. Its boundary of applicability is given by the statement: once the bottom-two step is added, elimination and combined score no longer correspond one-to-one and the criterion must be restated in two parts; moreover, the statement notes the existence of no-elimination weeks and of weeks in which more than one couple is eliminated, and such weeks cannot directly use the single “lowest” constraint.

Assumption 2 This assumption is a necessary condition for the identifiability of the estimate. If one takes only the single constraint that “the couple eliminated that week has the lowest combined score,” then each week pins down just one inequality, the vote shares satisfying it form a set far larger than a single point, and a point estimate is meaningless. Adding share normalization and across-week continuity narrows the feasible region substantially. The cost of this assumption is the introduction of a smoothness strength, a quantity that must be set on one’s own, whose value is discussed in the sensitivity section.

Assumption 3 This assumption underpins the recomputation of both combination rules. The statement notes that the decimals in the data are the average over multiple dances that week, that bonus points have been allocated in, and that team-dance scores have been averaged with individual scores, so the sum of a couple’s judge scores that week can serve as a single proxy for that couple’s performance that week. If the assumption fails, the results obtained by combining by percent would be mixed with differences in the data’s basis, and the comparison of the two rules would no longer be a comparison between the rules themselves.

Assumption 4 This assumption is a precondition for reading in the data, not a modeling assumption. On checking, the dataset contains 4741 cells of N/A and 4671 cells of 0; the former are concentrated in the column of the fourth judge and in the weeks not aired in each season, the latter in the rows of the eliminated couple for the remaining weeks of each season, matching the statement item by item.

Assumption 5 This assumption is one the statement explicitly marks as uncertain: the statement’s own wording says that the exact season of the return to the rank method cannot be verified, and that assuming it to be season 28 is reasonable. This paper adopts the value given by this statement; if the assumption fails, the combination rule of the seasons involved would be used wrongly, and their estimates and results would fail together, so its sensitivity is tested separately in Section 6.

3 Analysis of the Problem

The audience votes are the one quantity the dataset marks as confidential, yet every later question takes them as an input: part (a) asks for an estimate of each couple’s vote in each week it competed, and parts (b), (c) and (d) all treat that estimate as known. This section therefore settles where the unobservable quantity is to come from.

3.1 Three Candidate Directions

The first is to seek an external proxy. The statement permits supplementing the data, and audience votes are intuitively tied to social buzz or search trends, but no published conversion turns buzz into a vote share. A proxy can at best give an ordering, and whether that ordering transfers to votes is itself an empirical question rather than a definition — we take it up in Section 7 and report there what happened when we tried it.

The second is to invert from the elimination results. They are available week by week and season by season, and they are linked to the unknown votes directly through the season’s combination rule: the couple eliminated that week is the one with the lowest combined score. That sentence becomes an inequality about the unknown vote shares, and recording it week by week imposes constraints on them.

The third is to bypass the vote counts and compare only the rules. Each rule’s combined score is half fan vote share, so without the votes both rules stay at the level of their verbal definitions: neither who was eliminated nor whether the two rules differ can be computed.

3.2 Why the Second Direction

Three reasons support the second direction. Its observations come from the problem’s own attachment, so the estimation itself introduces no external data and no extra layer of data-sourcing or basis alignment; the external route is not abandoned but deferred — Section 7 reports the proxy we did obtain and why the records reject the ordering it implies. It matches the form of the information the statement supplies: the two worked examples in its appendix demonstrate precisely that “the eliminated couple is the one with the lowest combined score,” and the direction merely generalizes that criterion to all weeks. And it binds estimation to testing—the inverted shares are fed straight back into the season’s rule to recompute the week’s elimination, whose hit rate falls out exactly as the “measure of consistency with the elimination results” the statement requires.

3.3 Two Kinds of Structure the Inversion Must Supply

With only “the eliminated couple has the lowest combined score,” the model is underdetermined: each week pins down a single inequality while the number of unknowns equals the number of couples present that week. Two kinds of structure must therefore be supplied. The first is normalization: the vote shares of the couples present in a week sum to one. The second is across-week persistence:

the latent popularity of a couple changes continuously between adjacent weeks. The former confines the unknowns to a simplex; the latter compresses the week-by-week unknowns into a single quantity shared within a season. Their costs were argued in Assumption 2 of Section 2; here it suffices that they are the premise for the inversion to be solvable.

3.4 The Resulting Model Form and Its Connections

The model should therefore not be built as a point estimate, but as “a feasible set under constraints, together with a rule for drawing a point estimate from that set.” The two rulers part (a) asks for each get their place: certainty is the interval width of the feasible set, which varies with couple and with week; consistency is the hit rate of feeding the estimate back into the rule. Section 4 turns the three structures above into linear constraints, Section 5 reports the solution results, and Section 6 examines how far the conventions chosen here drive the conclusions.

4 Model Development

4.1 Inversion Framework: Constraining Vote Shares from Elimination Records

This section builds the model required by question (a) of the statement. The audience vote shares $v_{i,s,w}$ are unobservable; what is observable is each season’s weekly elimination results; the only information linking the two is the combination rule adopted that season (Assumptions 1 and 5). The modeling therefore proceeds in the inversion direction: “the couple eliminated that week has the lowest combined score” is written as a constraint on the unknown vote shares, and the estimate and the certainty measure are then read off the solution set of the constraints.

The observational unit is the triple (i, s, w) , i.e., couple i in week w of season s . The couples with a non-N/A, non-0 judge score that week are counted in the present set $A_{s,w}$ (Assumption 4); the couple eliminated that week is denoted $e_{s,w}$. The total judge score that week $J_{i,s,w}$ is the sum of the individual judge scores, and its share is denoted

$$j_{i,s,w} = J_{i,s,w} / \sum_{k \in A_{s,w}} J_{k,s,w}, \quad i \in A_{s,w}, \quad (1)$$

consistent with the percentage method given in the statement’s appendix. When combining by the percentage method, the eliminated couple must satisfy, for every contestant still present,

$$j_{e_{s,w},s,w} + v_{e_{s,w},s,w} \leq j_{j,s,w} + v_{j,s,w}, \quad j \in A_{s,w}, \quad j \neq e_{s,w}. \quad (2)$$

The vote shares are also subject to two prior constraints: non-negativity, and summing to one over those present that week (Assumption 2):

$$v_{i,s,w} \geq 0, \quad \sum_{i \in A_{s,w}} v_{i,s,w} = 1. \quad (3)$$

The motivation for Equation (2) and Equation (3) is direct. The audience votes are a set of non-negative shares whose weekly total is normalized; and a single elimination tells the model only “who is ultimately lowest,” without directly giving the ordering of each contestant’s vote count. Together the two characterize the set of vote shares satisfying the elimination records as the intersection of a simplex and several half-spaces, i.e., a polyhedron, and the solvability of the model comes precisely

from this geometric structure. Its cost is that the information is very weak: a single week yields at most a family of inequalities with a consistent direction, not enough to locate each contestant's share.

The rank-method seasons lie outside the definition of the above feasible region. The rank method orders couples by the sum of judge rank and vote rank; its constraint is an ordinal relation between ranks rather than a linear inequality in the vote shares, and so it does not form a polyhedron; this paper does not define a vote-share feasible region for the rank-method seasons. Of the four controversial cases named in the statement, seasons 4 and 11 fall within scope, while seasons 2 and 27 are rank-method seasons and lie outside scope.

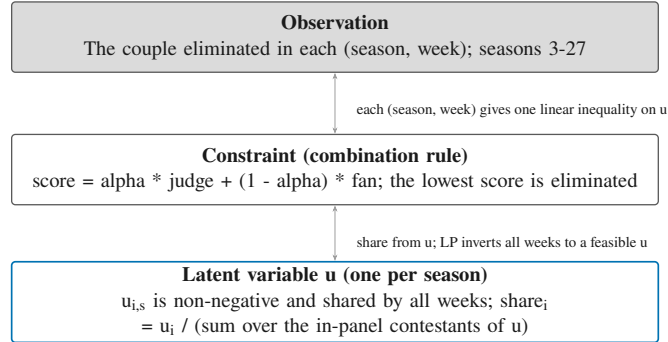


Figure 2: Structure of the inversion. The elimination record (observed) constrains the within-season latent popularity u (unobserved) through one linear inequality per (season, week); weekly vote shares are recovered as $u_i / \sum_{j \in A_w} u_j$.

4.2 Across-Week Smoothing: Within-Season Latent Popularity

To narrow the feasible region, the model writes in the across-week continuity of Assumption 2. Let couple i have a within-season-constant latent popularity $u_{i,s} \geq 0$ in season s , normalized to one over all contestants C_s of that season:

$$\sum_{i \in C_s} u_{i,s} = 1. \quad (4)$$

The weekly vote share is a relative quantity: within a single week, what determines rank is relative popularity. Extracting the absolute strength into the within-season invariant u , the weekly share is the relative popularity among those present that week,

$$v_{i,s,w} = u_{i,s} / \sum_{k \in A_{s,w}} u_{k,s}, \quad i \in A_{s,w}. \quad (5)$$

Write $S_{s,w} = \sum_{k \in A_{s,w}} u_{k,s}$, which is a linear form in u . Substituting Equation (5) into Equation (2) and clearing the denominator gives

$$u_{e_{s,w},s} - u_{j,s} \leq (j_{j,s,w} - j_{e_{s,w},s,w}) S_{s,w}, \quad (6)$$

whose right-hand side is the product of a constant and a linear form in u , so the constraint remains linear in u . The whole-season feasible region therefore remains a polyhedron, and finding the per-contestant bounds is still a linear program.

The model's adaptivity lies in this: the contestant set C_s , the present set $A_{s,w}$, and the judge shares j are all given by the local data, so a different dataset means a different set of constraints; the dimension of u equals the number of contestants in that season. The smoothness strength takes its strongest form, i.e., constant within a season, so the main model contains no additional free parameter; the strength itself, as a relaxable setting, is left to the sensitivity section.

4.3 Certainty Measure and Point Estimate

The certainty measure is given by per-contestant linear programming. For each $i \in C_s$, minimize and maximize $u_{i,s}$ under the constraints Equation (4) through Equation (6), obtaining the feasible interval $[\underline{u}_{i,s}, \bar{u}_{i,s}]$; the interval width $\bar{u}_{i,s} - \underline{u}_{i,s}$ is that couple's certainty measure, a narrower width meaning that the season's elimination records constrain that couple's popularity more tightly. The second part of question (a), "whether certainty varies from one couple to another," is in the model the question "whether these widths are constant."

The point estimate takes the midpoint of each interval and renormalizes to one, i.e., $u_{i,s}^* = (\underline{u}_{i,s} + \bar{u}_{i,s})/2$ followed by division by the sum over that period. This rule is deliberately simple: no random source, reproducible contestant by contestant. Its cost is that the midpoint need not lie within the feasible region, so the model additionally reports an upper bound—using all weeks' constraints to determine whether any feasible point exists that reproduces the elimination sequence of every week of that season; this upper bound is called the oracle below. If even the oracle cannot do it, then the criterion "elimination means lowest combined score" itself does not hold for the season under study (the boundary of Assumption 1).

Given u^* , the forward replay computes the vote shares week by week via Equation (5), then, via Equation (2), takes the couple with the lowest combined share as the predicted eliminated couple, for comparison with the observations.

4.4 Consistency Measure: Leave-One-Week-Out

The consistency measure uses the leave-one-week-out method. For week k of a season, fit the midpoint estimate u^* using only the elimination constraints for $w \leq k - 1$, then predict the eliminated couple of week k and compare with the true $e_{s,w}$; the hit rate is the number of hit weeks divided by the number of weeks involved in the prediction. Predictions are made only for weeks with an elimination record; no-elimination weeks and weeks in which more than one couple is eliminated do not participate (Assumption 1 has already explained that these two kinds of weeks cannot directly use the single "lowest" constraint).

4.5 Recomputation of the Two Combination Rules

Question (b) requires recomputing the two rules on each season. The percentage method's replay is Equation (2): eliminate, week by week, the couple with the lowest combined share, and remove it from the present set after elimination. The rank method's replay additionally denotes by R_i^J the couple's judge-score rank that week and by R_i^V its vote-share rank that week, both starting from 1 with 1 the highest; the combined quantity is the rank sum $R_i^J + R_i^V$, and the couple with the largest rank sum is eliminated week by week. A divergence between the two methods in the same week means the two methods give different eliminated couples; on the question "which method better favors the fan vote," the model takes the rank correlation of survival weeks with the fan vote and with the judge score, respectively, as its readings, making only correlational statements and not claiming causation.

4.6 Effects Model

The observational unit of question (c) is (season, couple). Response one $\bar{j}_{i,s}$ is the mean of the couple’s judge shares (Equation (1)) over the weeks it was present that season, and response two is $u_{i,s}^*$. The explanatory variables are age at the time of competing age, US-born-or-not is_us, industry dummies, and professional-dancer dummies (dummies are set only for dancers appearing in at least five weeks, the rest being pooled into the baseline); a season fixed effect is also included, so that between-season differences are not attributed to contestant characteristics. Estimation uses least squares with HC1 robust standard errors. Two models are fitted separately: M_1 contains only contestant characteristics and the season fixed effect, and M_2 adds the dancer fixed effect on top of that; each is fitted once for each of the two responses, and comparing the sign of a given variable’s coefficient across the two responses answers “whether the effects on judge score and on fan vote run in the same direction.”

4.7 Rule Family and the Bottom-Two Step

Question (d) searches over rule space. The rule family is taken as

$$m_i = \alpha j_i + (1 - \alpha) v_i, \quad \alpha \in \{0, 0.25, 0.5, 0.75, 1\}, \quad (7)$$

where $\alpha = 0.5$ is the current percentage method, $\alpha = 1$ looks only at the judges, and $\alpha = 0$ looks only at the fans. A switch b is also set: when on, the two couples with the lowest combined quantity m_i are first taken, and the judges then save the one with the higher judge score, the other being eliminated—which is the formalization of the step added from season 28 onward. The statement says only that this step is decided by a judges’ vote and does not say how the judges vote, so the model assumes the judges keep the higher-scoring couple; this is a replaceable modeling assumption, and replacing it would change the values of the fairness quantities below. Evaluation uses two objectives: fairness O_1 is the rank correlation of the elimination order with the judge-score ordering, and fan engagement O_2 is the rank correlation of the elimination order with the vote-share ordering. The two point in opposite directions, forming the fairness-versus-interest trade-off frontier.

5 Solution and Results

5.1 Solution Procedure

The units of solution and the notation for the four steps below follow the previous section. Inversion and certainty are carried out by linear programming: for each (season, week) with an elimination record, and for each season, the HiGHS solver minimizes and maximizes the within-season latent popularity u for each contestant in turn, giving a feasible interval; the point estimate is the interval midpoint, renormalized to one. Consistency uses the leave-one-week-out refit, week by week. The rule replays follow the definitions of the previous section, week by week, where both the percentage method and the rank method are compared on the same true field of that week, so that after the first divergence the two are not comparing two different populations. The effect model is least squares with robust standard errors, and the rule-family sweep is a replay group by group. All computation is a deterministic pipeline with a fixed random seed (2025) at the entry point so that it reproduces exactly; no sampling is used in this batch of computation. The correspondence between each artifact and its generating script is given in the results list.

5.2 Problem (a): Feasible Region, Certainty and Consistency

The feasible region is non-empty. Among the weeks with an elimination record, the single-week level has zero infeasible cases, i.e. 0/197 weeks (of the 198 weeks with an elimination record, 1 week is skipped because fewer than two contestants are present that week); the whole-season level has zero infeasible cases, i.e. 0/25 seasons. The first falsification condition listed at the model-selection stage—that the feasible region is generally empty—does not hold.

Certainty is measured by the feasible-interval width (Tables 2 and 3). Under single-week constraints the median width is 0.992 and P_{90} is 1.000, almost the whole set from 0 to 1; once cross-week smoothing is imposed, the season-wide median width falls to 0.1622 and P_{90} falls to 0.9301. Figure 3 gives the empirical distribution of the width before and after smoothing: cross-week smoothing presses the median width down to about one sixth, but P_{90} is still 0.9301, so about one tenth of the contestants remain nearly unidentifiable under the season-wide constraints.

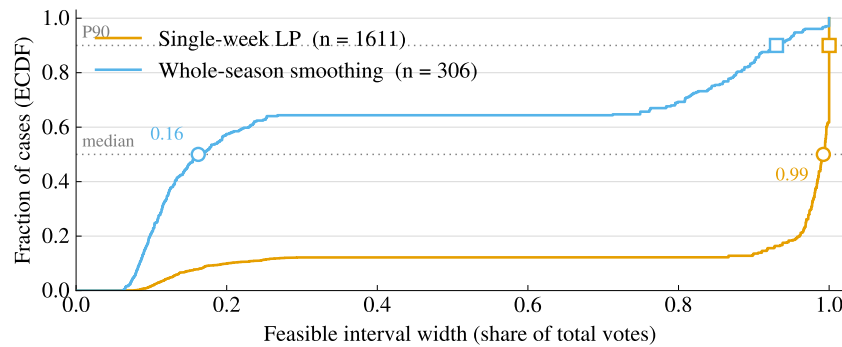


Figure 3: Empirical cumulative distribution of the feasible-interval width, per contestant-week under single-week constraints ($n = 1611$) versus per contestant under the whole-season latent variable ($n = 306$). Median and 90th percentile are marked. Cross-week pooling shrinks the median from 0.99 to 0.16, but the 90th percentile stays at 0.93.

Certainty varies from contestant to contestant. Figure 4 gives the feasible interval of each contestant of season 25: a wide one such as Barbara Corcoran reaches 0.92, nearly filling 0 to 1, while a narrow one such as Debbie Gibson is only 0.06. Figure 5 gives the season-by-season mean width over the 25 seasons, in which season 19 is the most identifiable (mean 0.25) and season 26 the least (0.70, and that season ran only 3 weeks). Hence the answer to the second question of part (a): certainty is not a constant but varies with the contestant and the week.

Consistency is shown in Table 4. The leave-one-week-out procedure makes 172 predictions in all and hits 67, giving a consistency rate of 0.390. The random baseline is about 0.10, and the oracle upper bound is 1.000, i.e. within the season-wide feasible region there exists a solution that reproduces the entire elimination sequence of that season. That is, the criterion “the lowest combined score is eliminated” is satisfiable on the season-wide scale, and the shortfall of 0.390 comes from the point-estimate convention itself, not from the criterion being unsatisfiable.

5.3 Problem (b): Comparison of the Two Combination Rules

See Table 5. The week-by-week replay hit rate of the percentage method is 0.505, i.e. 100/198. The proportion of weeks on which the two methods give a different eliminated contestant is 0.197, i.e. 39/198 weeks. So the choice of rule does change the outcome, but a flip occurs in only about one fifth of the weeks.

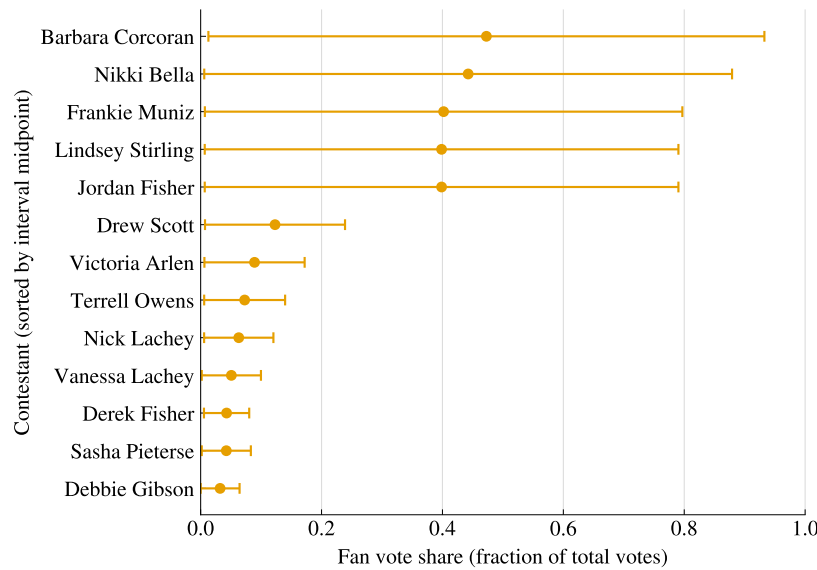


Figure 4: Feasible interval of the season-level latent fan-vote share for every contestant of season 25, sorted by interval midpoint. Widths range from 0.06 (Debbie Gibson) to 0.92 (Barbara Corcoran): the certainty of the estimate is not the same for all contestants.

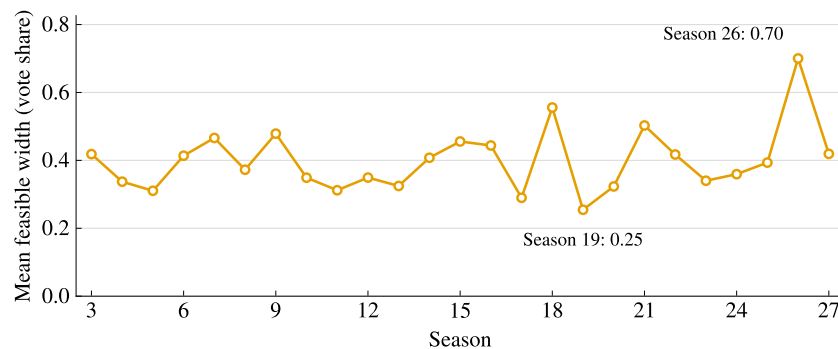


Figure 5: Mean feasible-interval width by season, over the percentage-method seasons 3–27. Season 19 is the most identifiable (0.25) and season 26 the least (0.70; that season contributes only three replayable weeks).

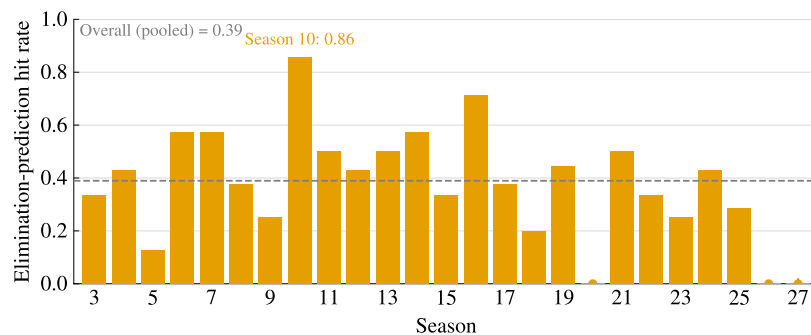


Figure 6: Leave-one-week-out hit rate by season, with the overall rate 0.390 marked. Zero-hit seasons cluster in the short seasons, where few weeks are available to fit.

Table 2: Feasible interval widths of the per-week vote share, aggregated by season. Columns: number of contestant-week intervals; minimum, median and maximum interval width (unit: share of the weekly total). The *Total* row pools all seasons.

Season	n	min	median	max	Season	n	min	median	max
3	54	0.104	0.974	1.000	16	60	0.118	1.000	1.000
4	60	0.096	0.984	1.000	17	72	0.099	0.989	1.000
5	72	0.094	0.996	1.000	18	39	0.117	0.981	1.000
6	61	0.101	0.991	1.000	19	85	0.089	0.989	1.000
7	63	0.105	1.000	1.000	20	67	0.077	0.981	1.000
8	75	0.086	0.977	1.000	21	57	0.098	1.000	1.000
9	89	0.078	0.992	1.000	22	61	0.106	0.998	1.000
10	60	0.091	0.997	1.000	23	75	0.086	0.994	1.000
11	72	0.101	1.000	1.000	24	66	0.100	1.000	1.000
12	60	0.103	1.000	1.000	25	66	0.079	0.978	1.000
13	72	0.088	0.995	1.000	26	24	0.114	0.962	0.995
14	67	0.096	1.000	1.000	27	69	0.085	0.986	0.997
15	65	0.085	0.994	1.000	Total	1611	0.077	0.992	1.000

Intervals by linear programming per (season, week), scoped to percent-rule seasons 3–27.

Table 3: Season-level latent fan-vote bounds (seasons 3–27). Columns: contestants with a latent variable n ; elimination weeks replayable; linear constraints in the season program; mean interval width. The *Total* row reports column sums, except the last column (overall mean width).

Season	People	Weeks	Cons.	Mean width	Season	People	Weeks	Cons.	Mean width
3	11	7	47	0.4184	16	12	8	52	0.4438
4	11	8	52	0.3373	17	12	9	63	0.2899
5	12	9	63	0.3104	18	12	6	33	0.5558
6	12	8	53	0.4137	19	13	10	75	0.2546
7	13	8	55	0.4660	20	12	8	59	0.3232
8	13	9	66	0.3724	21	13	7	50	0.5027
9	16	9	80	0.4788	22	12	7	54	0.4173
10	11	8	52	0.3491	23	13	9	66	0.3400
11	12	9	63	0.3120	24	12	8	58	0.3594
12	11	8	52	0.3494	25	13	8	58	0.3934
13	12	9	63	0.3246	26	10	3	21	0.7003
14	12	8	59	0.4077	27	13	7	62	0.4192
15	13	7	58	0.4557	Total	306	197	1414	0.3994

A single latent share per contestant per season, fitted to all weekly constraints.

Table 4: Leave-one-week-out consistency of the predicted eliminee, by season. Columns: number of weekly predictions made; number of hits; hit rate. The prediction uses the midpoint of the feasible interval as the fan-vote estimate.

Season	Pred.	Hits	Rate	Season	Pred.	Hits	Rate
3	6	2	0.333	16	7	5	0.714
4	7	3	0.429	17	8	3	0.375
5	8	1	0.125	18	5	1	0.200
6	7	4	0.571	19	9	4	0.444
7	7	4	0.571	20	7	0	0.000
8	8	3	0.375	21	6	3	0.500
9	8	2	0.250	22	6	2	0.333
10	7	6	0.857	23	8	2	0.250
11	8	4	0.500	24	7	3	0.429
12	7	3	0.429	25	7	2	0.286
13	8	4	0.500	26	2	0	0.000
14	7	4	0.571	27	6	0	0.000
15	6	2	0.333	Total	172	67	0.390

Random baseline ≈ 0.10 , oracle = 1.000.

Table 5: Percent vs. rank rule, replayed season by season. Columns: weeks replayed; eliminations reproduced by the percent rule; weeks where the two rules disagree; survival-share correlations (Spearman) with fan and judge shares under each rule; in-scope controversy cases. “--” = not defined for that season.

Season	Weeks	Hits	Disagr.	$r_{\text{pct}}^{\text{fan}}$	$r_{\text{rk}}^{\text{fan}}$	$r_{\text{pct}}^{\text{jud}}$	$r_{\text{rk}}^{\text{jud}}$	Controversy
3	7	2	2	0.9643	0.9643	0.9643	0.9643	–
4	8	2	2	0.9048	0.8571	0.9048	0.9286	Billy Ray Cyrus
5	9	1	0	0.8833	0.9667	0.9833	0.9833	–
6	8	8	1	0.9762	0.9762	1.0000	1.0000	–
7	8	8	0	0.9286	0.9286	0.9762	0.9762	–
8	9	3	1	0.9333	0.9333	0.8333	0.8833	–
9	9	3	2	0.9167	0.9500	1.0000	0.9667	–
10	8	3	3	0.4762	0.6429	1.0000	0.9286	–
11	9	6	0	0.9167	0.9167	0.9667	0.9833	Bristol Palin
12	8	8	3	0.9762	0.9762	0.9762	0.9762	–
13	9	4	3	0.9000	0.9333	0.9667	0.9500	–
14	8	5	3	0.9762	0.9762	0.9048	0.9762	–
15	7	3	2	0.8214	0.9643	1.0000	1.0000	–
16	8	8	0	1.0000	1.0000	0.9524	0.9524	–
17	9	9	3	0.9500	0.9500	0.9667	0.9667	–
18	7	1	2	0.9643	0.9286	1.0000	1.0000	–
19	10	3	1	0.9758	0.9515	0.9879	0.9879	–
20	8	0	1	0.9524	0.9048	0.9762	1.0000	–
21	7	7	1	0.8929	0.8929	1.0000	1.0000	–
22	7	2	2	0.9286	1.0000	1.0000	0.9643	–
23	9	2	2	0.9500	0.9000	1.0000	1.0000	–
24	8	4	2	0.9762	0.9762	1.0000	1.0000	–
25	8	0	2	0.9524	0.8810	0.9762	1.0000	–
26	3	3	0	–	–	–	–	–
27	7	5	1	0.8929	0.8929	0.7500	0.7857	–

Scope = percent-rule seasons 3–27, season 26 replayed only 3 weeks.

The fan leaning is measured by rank correlation. The rank correlation of survivor-weeks with the fan vote is 0.917 under the percentage method and 0.928 under the rank method; the rank method is higher in only 6/24 seasons. The two methods differ sporadically from week to week and share almost the same long-run correlation, so there is no evidence that either method systematically favors the fan vote; this item is an association statement only.

Of the controversial cases named in the problem, the two that fall inside the scope give these results: in season 4 (Billy Ray Cyrus) the two methods disagree on 2 weeks, and in season 11 (Bristol Palin) they disagree on 0 weeks. Changing the combination rule does not always flip the outcome. Season 2 is a rank-method season and is not covered by this step; the week-by-week eliminations of season 27 are inside the scope of this step, but its controversy concerns the win, which is decided by the final placing and is not an object that week-by-week eliminations can characterize.

5.4 Problem (c): Effects of Dancer and Contestant Characteristics

See Table 6. The coefficients of age on the judge share and the fan vote are -0.0012 (robust standard error 0.0001) and -0.0015 (0.0003), both negative and both many times their standard errors. The US-born indicator is -0.0056 (0.0060) and -0.0156 (0.0124), same-signed but both insignificant. After adding dancer fixed effects, the age coefficient moves only from -0.0012 to -0.0010 , showing that the dancer fixed effects absorb very little. Both responses are shares, so the coefficient units are “share per year of age” and “share”. Hence the answer to part (c): the two variables act on the judge score and the fan vote in the same direction.

Table 6: Effects of contestant features on judge and fan scores (OLS; each coefficient is followed by its standard error in the adjacent “s.e.” column). Same = Yes when the two coefficients share a sign.

Model	Var.	Judge coef	s.e.	Fan coef	s.e.	Same
M1 (contestant)	age	-0.0012	0.0001	-0.0015	0.0003	Yes
M1 (contestant)	is_us	-0.0056	0.0060	-0.0156	0.0124	Yes
M2 (contestant + dancer FE)	age	-0.0010	0.0002	-0.0014	0.0004	Yes
M2 (contestant + dancer FE)	is_us	-0.0036	0.0057	-0.0128	0.0136	Yes

M2 adds dancer fixed effects.

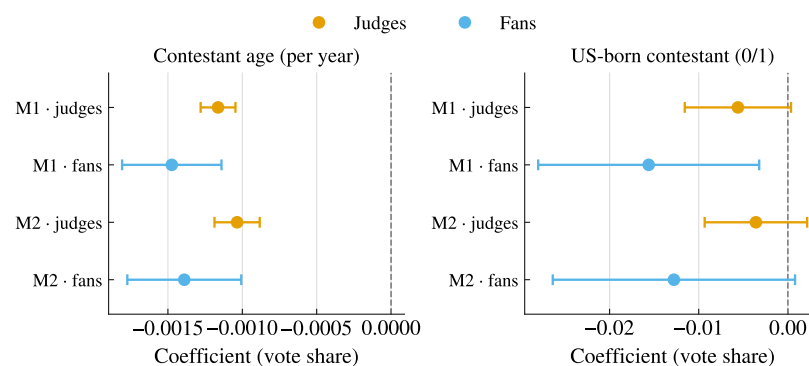


Figure 7: Estimated effects of contestant characteristics on the judge share and on the fan share, each with its robust standard error, split by variable to avoid a tenfold scale difference. Both responses move in the same direction.

5.5 Problem (d): Mechanism Design

See Table 7. The judge consistency O_1 and the fan consistency O_2 of the current rule ($\alpha = 0.5$, no bottom-two step) are 0.9619 and 0.9170. The recommended combination $\alpha \approx 0.25$ together with the bottom-two step gives $O_1 = 0.9746$ and $O_2 = 0.9175$: judge consistency is clearly better than the current rule ($0.9746 > 0.9619$) and fan consistency is no worse ($0.9175 \geq 0.9170$). As a control, $\alpha = 0.75$ with the bottom-two step gives $O_1 = 0.9929$ and $O_2 = 0.8212$, and at $\alpha = 1.00$ O_1 rises to 0.9950 while O_2 falls to 0.5891, showing that raising the judge weight clearly sacrifices fan consistency. The judge-consistency values above rely on the assumption “the judges keep the higher-scoring contestant in the bottom-two step”: the direction of the conclusion (adding the bottom-two step raises judge consistency) is insensitive to this assumption, while the magnitudes are sensitive.

Table 7: Mechanism design: combined rank $C_i = \alpha \text{ judge}_i + (1 - \alpha) \text{ fan}_i$ with and without a bottom-two step. O1 = judge-score consistency; O2 = fan-share consistency; replayed over 24 seasons.

α	Bottom-2	O1 judge	O2 fan	Seasons
0.00	No	0.9347	1.0000	24
0.00	Yes	0.9665	0.9664	24
0.25	No	0.9316	0.9353	24
0.25	Yes	0.9746	0.9175	24
0.50	No	0.9619	0.9170	24
0.50	Yes	0.9813	0.8777	24
0.75	No	0.9866	0.8546	24
0.75	Yes	0.9929	0.8212	24
1.00	No	0.9950	0.5891	24
1.00	Yes	0.9943	0.5959	24

$\alpha = 0.50$ without bottom-2 is the current rule; recommended $\alpha \approx 0.25$ with bottom-2.

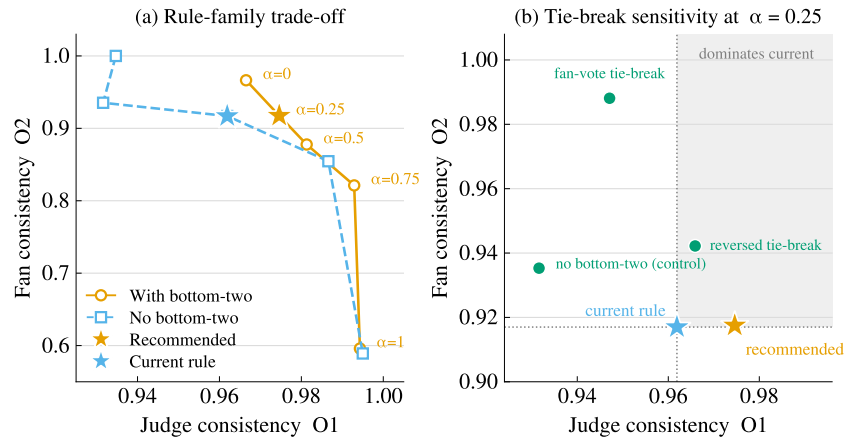


Figure 8: The rule family traced out in the (fairness, engagement) plane. Panel (a) shows the frontier as the judge weight α rises; panel (b) places the recommended rule ($\alpha \approx 0.25$ with a bottom-two step) against the incumbent, and marks the variants of the bottom-two tie-break, on which the domination claim depends.

5.6 Scope of Applicability of the Results

The following three conventions must be read together with the readings above; this section does not claim to have listed the scope of applicability exhaustively.

First, the scope. All computation covers only the percentage-method seasons (seasons 3 to 27, 25 seasons in all); the rank-method seasons (seasons 1 to 2, and season 28 onward) are not covered, because the inversion step has no feasible region of vote shares for them—their constraint is the order of the rank sums, not a linear inequality on shares. Any solution that relies on vote shares is confined to the percentage-method seasons. Of the four controversial cases named in the problem, season 2 lies outside the scope.

Second, the meaning of 0.390. It is the consistency rate achieved by a deliberately simple point estimate (the feasible-interval midpoint) and does not represent an upper bound for this model class; the oracle upper bound under the same model is 1.000.

Third, the meaning of u . $u_{i,s}$ is the latent variable of within-season latent popularity, not the actual number of votes in any week; wherever parts (b), (c) and (d) use u as the fan vote, they are using this estimator, whose uncertainty is measured by the feasible intervals given in part (a).

6 Sensitivity, Error and Stability

This section asks two things: where the uncertainty of the estimator comes from, and how far the conclusions depend on the conventions we chose ourselves during modelling. Each of three conventions is changed in turn and the same batch of readings is rerun; the script is `sensitivity.py`, with entry seed 2025, and the readings are given in Table 8.

6.1 Sources of Error

The error in this problem has two layers, which must not be conflated. The first is in the data: judge scores are recorded to a precision of 0.5, the scores of several dances in a week are averaged, and bonus points are spread into the individual judge scores; the model simply takes the sum of the judge scores as the week’s performance proxy, with no separate weighting. A few weeks do not enter the computation at all—no-elimination weeks and weeks with several eliminations do not fit the single constraint “the lowest is eliminated”, and the final-week placing lies outside the week-by-week convention. The second layer is in the estimator: the within-season latent popularity is a hidden variable, not the actual vote share of any week, but the product of pressing the week-by-week share constraints into one cross-week shared quantity. Its uncertainty is expressed by the feasible-interval width of Section 5, obtained by no repeated sampling. The point estimate is the interval midpoint, which need not lie inside the feasible region—a deliberate trade-off between “exactly reproducible” and “necessarily a feasible solution”. One further convention moves the readings: the observation does not resolve ties for the lowest combined score. Taking the first in traversal order gives a consistency rate of 0.390; requiring a unique lowest falls to 0.361; counting any of the tied as a hit rises to 0.401. The rate thus swings by about 0.04 with “how ties are decided”.

6.2 Sensitivity I: The Implementation Convention for Cross-Week Smoothing

Assumption 2 presses the week-by-week unknown into a within-season shared quantity, but “which point to take from the feasible set” is a separate and changeable convention. Taking each contestant’s interval midpoint and renormalizing gives the 0.390 of Section 5. Normalizing by the

lower bound of the interval lowers it to 0.128; by the upper bound, 0.378; drawing uniformly at random inside the feasible region puts it between 0.128 and 0.180, with a mean of 0.145 over two hundred draws. Two things follow: the rate is highly sensitive to how the point estimate is taken, so 0.390 is the current convention's achievement and must not be read as a capability ceiling of this model class; and among the four conventions tested the midpoint comes first, so the current convention is not an arbitrary choice.

6.3 Sensitivity II: The Season Boundary of the Combination Rule

Assumption 5 takes seasons 3 to 27 as the percentage-method seasons. The problem labels this stretch 27a and states in a footnote that the exact season of the return to the rank method cannot be verified, so the boundary is itself uncertain. Shifting it left by one and by two seasons moves the consistency rate from 0.390 to 0.404 and 0.409, the two rules' disagreement rate from 0.197 to 0.199 and 0.202, and the percentage method's weekly-replay hit rate from 0.505 to 0.497 and 0.489. All three move only at the level of one percent, and the direction of the part (b) conclusion does not change. Shifting the boundary right (season 28 onward) cannot be tested the same way: this model has no feasible region for the rank-method seasons, and testing it would require first changing the very assumption under test.

Table 8: Sensitivity of the reported readings to three modelling choices and to one numerical convention. **A:** point-estimate convention. **B:** season-to-rule boundary of Assumption 5. **C:** tie-break rule inside the bottom-two step, at two judge weights α ; the incumbent rule is $\alpha = 0.5$ without a bottom-two step. **D:** treatment of ties for the lowest combined score.

Panel A — Point estimate		Consistency			
Midpoint (current)		0.390	Panel D — Ties (equal lowest score)		
Lower bound, normalized		0.128	Keep the earliest (current)		0.390
Upper bound, normalized		0.378	Require a unique lowest		0.361
Random feasible point		0.145	Allow ties		0.401
(200 draws; range 0.128–0.180)					

Panel B — Season-to-rule boundary (Assumption 5)	Seasons	Consistency	Replay hit rate	Disagreement rate
Seasons 3–27 (current assumption)	25	0.390	0.505	0.197
Seasons 3–26	24	0.404	0.497	0.199
Seasons 3–25	23	0.409	0.489	0.202

Panel C — Bottom-two tie-break ($\alpha = 0.25$)	O_1 fairness	O_2 fan
No bottom-two step (control)	0.9316	0.9353
Judges keep the higher score (current assumption)	0.9746	0.9175
Judges keep the lower score (reversed)	0.9659	0.9422
Judges keep the higher fan vote	0.9471	0.9881
Control: current rule $\alpha = 0.5$, no bottom-two step	0.9619	0.9170

6.4 Sensitivity III: The Tie-Break Rule of the Bottom-Two Step

In the rule family of part (d), “the judges keep the higher-scoring contestant” in the bottom-two step is our assumption—the problem says only that a judges’ vote chooses one of the two, and does not say on what basis the judges vote. Rerunning with two equally defensible rules: with the judges keeping the higher score, adding the bottom-two step raises judge consistency from 0.9316 to 0.9746 at judge weight 0.25; keeping the higher fan vote, it still rises, to 0.9471; keeping the lower score, to 0.9659. Under all three the direction “adding the bottom-two step raises judge consistency” holds, with magnitudes differing by at most 0.028. The claimed dominance over the current rule is more fragile: the current rule (judge weight 0.5, no bottom-two step) reads 0.9619, and the recommended combination is 0.9746 under the same tie-break and 0.9659 under the reversed one, both above it, but only 0.9471 under tie-breaking on the fan vote, below the current 0.9619. The direction is thus insensitive to the assumption while the assertion “dominates the current rule” is sensitive to it, so this paper confines that assertion to the rule in which the judges tie-break on score.

6.5 Stability

At the numerical level, the inversion program is infeasible at no point over the 197 weeks that enter it, and the season-wide feasible region is non-empty over all 25 percentage-method seasons. Cross-week smoothing presses the median feasible-interval width from 0.992 for a single week to 0.1622 for a whole season, about six times, so the structure “cross-week persistence” carries most of the identifiability. The variation between seasons is pronounced—the season consistency rate runs from 0 (seasons 20, 26, 27) to 0.857 (season 10)—so a single overall rate does not describe the model’s performance. At the level of conclusions, the three directions—part (b) “no systematic leaning between the two methods”, part (c) “the characteristics act on the two responses in the same direction”, part (d) “adding the bottom-two step raises judge consistency”—are all unchanged under this section’s three perturbations.

7 Strengths and Weaknesses of the Model

This section evaluates the model built in Sections 4 and 5: under what conditions it is usable, and how far its scope reaches, item by item rather than as a list of impressions.

7.1 Strengths

7.1.1 The Constraints Come Only From the Problem’s Own Records

The inversion’s constraints come from the weekly elimination records and the season’s combination rule, so no extra uncertainty enters from data provenance or from aligning an external basis, and with the same data and the same seed the readings reproduce exactly. This is a property of what we did, not a claim that the problem needs no other data: the statement explicitly permits additional data provided the sources are documented, and the price of not using any is paid in identification — see the weakness below.

7.1.2 Certainty and Consistency Each Have a Quantity to Read Off

The feasible-interval width answers “how precise is this estimate” and varies with contestant and week; the replay hit rate answers “can the estimate reproduce the elimination outcomes”. Both rulers

part (a) asks for thus have a quantity rather than a qualitative description.

7.1.3 The Comparison of the Two Rules Reduces to the Rules Themselves

Both methods are applied to the same batch of estimates, and the switch of rule is a deterministic transformation on those estimates that introduces no second model, so “which method better favours the fan vote” is not contaminated by differences between estimation methods.

7.1.4 The Mechanism Design Yields a Segment of a Dominance Relation over the Rule Family

Part (d)’s answer is a comparison obtained by enumerating the judge weight and toggling the bottom-two step, reporting judge consistency and fan consistency at once, so the trade-off surface is visible.

7.2 Weaknesses

7.2.1 Scope Reaches Only the Percentage-Method Seasons

The rank-method seasons (1–2, and 28 onward) are constrained by the order of the rank sums, not by a linear inequality on vote shares, so the inversion has no feasible region for them and no part of (a)–(d) that relies on the latent variable covers them. Of the four controversial contestants named, Jerry Rice (season 2) is outside the scope and seasons 4 and 11 are inside it; season 27’s week-by-week eliminations are inside, but its controversy is on the final placing, which week-by-week eliminations do not decide.

7.2.2 The Consistency Rate Is Not an Upper Bound, and It Is Tied to the Point-Taking Convention

The 0.390 of Section 5 is the achievement of a deliberately simple point estimate, not an upper bound for this model class. Changing the point-estimate and tie-decision conventions in Section 6 leaves it between 0.128 and 0.401, while the same data reach 1.000 under the oracle convention: the shortfall is the estimator’s, not an unsatisfiable criterion’s, and the number must be cited together with its convention.

7.2.3 Within-Season Latent Popularity Is a Latent Variable

It presses the season’s share constraints into one quantity and is not any single week’s vote count. Wherever parts (b), (c) and (d) use it as the fan vote they use this estimator, whose uncertainty is the feasible intervals of Section 5; it cannot answer “how many votes a contestant actually received in a given week”.

7.2.4 The Season-to-Rule Mapping and the Bottom-Two Tie-Break Are Assumptions of Ours

This paper takes seasons 3 to 27 as percentage-method; if the true boundary is earlier, the corresponding seasons move out of the percentage set. Section 6 has rerun with the boundary shifted left by one and by two seasons and the directions are unchanged, but the item rests on a reasonable assumption the problem marks as uncertain, not on an established fact. Separately, the problem says only that one of the bottom two is chosen by a judges’ vote, not on what basis the judges vote: Section 6 shows the direction “adding the bottom-two step raises judge consistency” is insensitive to this,

while the item “the recommended combination dominates the current rule” does not hold if the judges are changed to tie-break on the fan vote.

7.2.5 An External Proxy Was Tried, and It Does Not Pay the Identification Price

Identification is the binding constraint on how sharp the estimate can be. Each week contributes exactly one inequality — the eliminated contestant must be the lowest combined — and one inequality per week is weak evidence: under single-week constraints the median interval width is 0.992, and even after the across-week pooling the 90th percentile is still 0.9301, so about one tenth of the contestants remain essentially unidentifiable. The statement allows additional data provided the sources are documented, and a popularity proxy is the single most promising addition, because it would enter as a second, independent signal on the very quantity the constraints leave loose.

We did obtain one — the daily page views of each contestant’s Wikipedia article over that season’s air weeks (Wikimedia Pageviews REST API) — and it does not pay that price. Read as an ordering on the latent popularity it is rejected by the records: seasons 24 and 27 admit no feasible point even at a 20-to-1 attention margin, and seasons 22 and 23 fail there too; where it is satisfiable at a weak margin it is binding rather than informative, collapsing the intervals to points in seasons 21 and 25. Read as an external ruler it agrees, correlating negatively with the estimate’s midpoint in five of the seven seasons covered — the only ones the endpoint holds at all, since it has nothing before 1 July 2015. We therefore report intervals rather than claims about individual weekly vote counts, and record the attempt rather than a conversion factor we do not have.

7.2.6 The Characteristic Analysis Claims Only Co-Direction, Not Causation

One regression response is itself an estimated latent variable, which pulls coefficients toward zero, so a statement of the kind “the older, the worse the performance” can be read only as association.

7.3 Conditions of Applicability

The model is more trustworthy on “rule comparisons within the percentage-method seasons, and the direction of mechanism improvement”: there the constraints are writable and replayable and the three conclusion directions survive all three perturbations. It is less trustworthy on “the specific vote count of a given contestant in a given week”: the median single-week interval width is 0.992, so a single-week constraint rules out almost no share combination and the point estimate is one representative taken from a very wide set.

8 Conclusion

This paper delivers a model that inverts fan votes from elimination outcomes, a procedure that compares the two combination rules on top of that estimate, and a set of improvement recommendations obtained over the rule family; a one- to two-page memo to the producers is attached separately. The conclusions below recycle the five parts of the problem, with all values taken from Sections 5 and 6.

First, the fan-vote estimate and its two rulers. The constraints built week by week and season by season from “the eliminated contestant of the week has the lowest combined score” are all feasible over the 197 weeks that enter the inversion in the percentage-method seasons, and the season-wide feasible region is non-empty over all 25 seasons. Certainty is the feasible-interval width: the median

single-week width is 0.992 and the season-wide median is 0.1622, cross-week smoothing pressing it down to about one sixth; the season-wide 90th percentile is 0.9301, about one tenth of the contestants remaining essentially unidentifiable even under the season-wide convention. Certainty is not the same for all—within a single season the narrowest contestant has an interval about 0.06 wide and the widest reaches 0.92 (see Figure 4)—so the answer to the question asked in the problem, “does certainty vary with the contestant and the week”, is yes. Consistency is the proportion of replay hits: leave-one-week-out makes 172 predictions and hits 67, a consistency rate of 0.390 against a random baseline of about 0.10.

Second, the two combination rules. Applied to each season, the two methods give a different eliminated contestant on 39 of 198 weeks, a disagreement rate of 0.197; the survivor–fan-vote rank correlation is almost identical under the two methods (percentage 0.917, rank 0.928), the rank method higher in only 6 of the 24 comparable seasons. There is thus no systematic leaning between the two methods, and the differences are sporadic. Of the controversial contestants named in the problem, the two inside the scope are season 4 (Billy Ray Cyrus), where the methods differ on two weeks, and season 11 (Bristol Palin), with no week of difference.

Third, the effects of the characteristics. With (season, contestant) as the observation unit and season fixed effects controlled for, the coefficients of age on the judge score and the fan vote are both negative (−0.0012 and −0.0015 with contestant characteristics only; −0.0010 and −0.0014 after adding dancer fixed effects), and the US-born coefficients are same-signed but insignificant. The signs agree across the two responses, so these characteristics act on the judge score and the fan vote in the same direction; the conclusion claims association only.

Fourth, the rule recommendation. Enumerating the rule family in which the combined score is the weighted sum of the judge share and the fan share, we recommend lowering the judge weight to about 0.25 and adding the bottom-two step. Judge consistency reads 0.9746 against the current 0.9619 and fan consistency 0.9175 against the current 0.9170: a clear improvement in judge consistency with no loss in fan consistency. This explains the direction of the change described in the problem, the addition of the bottom-two step in season 28, and indicates that it should be paired with a lower judge weight. Section 6 has shown that the dominance holds under the rule in which the judges tie-break on score and does not hold if the judges tie-break on the fan vote.

Fifth, the report and the memo. A complete report of no more than 25 pages and a one- to two-page memo to the producers accompany this paper, the memo summarizing the conclusions above and giving the same set of recommendations.

The conclusion of the whole paper can be condensed into one sentence: the current equal-weight sum puts the judges and the audience on the same weight, whereas the data show that lowering the judge weight to about 0.25 while letting the judges make one save decision among the bottom two raises the agreement of the outcome with professional judgment without sacrificing audience participation. Two things can be done next: build an inversion for the rank-method seasons that fits the rank-sum constraint, and feed samples inside the feasible region directly into the characteristic regression to avoid the coefficient attenuation that comes from using an estimator as a response. A third — using an external popularity signal to narrow the single-week intervals — we have since tried rather than merely proposed: Section 7 reports the proxy we obtained and why the elimination records reject the ordering it implies for every season the endpoint covers.

9 Memo (to the Producers of DWTS)

To: The producers of DWTS (*Dancing with the Stars*) **From:** Team #1111111 **Date:** October 9, 2026

Subject: How to combine judge scores and fan votes — conclusions, evidence, and recommendations for future seasons

Core recommendation. Combine judge scores and fan votes under the **percentage method**, lower the judge weight to about **one quarter** (write the judge weight in the combined score as α , i.e. $\alpha \approx 0.25$; fans take about three quarters), and **add a bottom-two step in which the judges decide the elimination** (the show has moved in this direction from season 28 onward; our results support this step and show that it should come with a lower judge weight). Under the two measures we use, this rule is **better on fairness than** the current equal-weight rule (judges and fans at half each, no bottom-two step) and **no worse on audience engagement. But this “better” has a precondition:** it holds because the **bottom-two step is decided by the judges, who keep the higher-scoring couple;** if the judges instead keep the higher **fan vote**, the advantage disappears (0.9471, below the current 0.9619). See Section 6 of the paper.

Evidence. We recovered the secret fan votes from the observable weekly eliminations, then replayed the same votes under each combination rule. Two findings. First, the rank method and the percentage method **do not systematically favor the fans** — the rank agreement of survivors with fan votes is 0.917 versus 0.928, nearly identical, and the difference is only a sporadic flip in about **one fifth** of the weeks (39/198). Second, the **only reliable** step that improves the outcome is a **bottom-two step decided by the judges**, paired with a lower judge weight.

Key numbers. We measure the outcome by two **rank correlations**, both **our own construction:** *fairness* = the agreement of the combined rank with the judge rank; *audience engagement* = the agreement of the combined rank with the fan-vote rank.

Combination rule	Fairness	Audience engagement
Current (equal weight): judges and fans at half each, no bottom-two step	0.9619	0.9170
Recommended: judges at 1/4 + bottom-two step	0.9746	0.9175
Judges at 3/4 + bottom-two step	0.9929	0.8212
Judges only (no fan votes)	0.9950	0.5891

Raising the judge weight further lifts *fairness* but *audience engagement* **collapses** (0.8212, 0.5891) — fans would feel their vote counts for nothing; changing nothing leaves the recommendation **dominating on fairness**.

How sure we are. The fan votes are secret, but each week’s “who was eliminated” pins them into an interval: a feasible vote-share distribution exists in all 197 decidable weeks and all 25 seasons. Cross-week smoothing narrows the interval from about 0.99 in a single week to a season-wide median of about 0.16, and about **one contestant in ten** remains essentially unidentifiable ($P_{90} = 0.93$). Under leave-one-out, our point estimate reproduces that week’s elimination at a rate of 0.390 (random baseline about 0.10, upper bound 1.000) — the record of a deliberately simple point estimate, not a ceiling.

Scope and two disclosures.

- **Reach:** all computation **covers only the percentage-method seasons (seasons 3–27)**. The rank-method seasons (seasons 1–2 and season 28 onward) are **not covered** by us (the inverse problem has no feasible region for the rank method), so the recommendation applies only to the percentage rule.
- **Measures:** *fairness* and *audience engagement* are **our own rank correlations**; we **do not claim** that the new rule is “fairer” as a general fact, and **claim only** that it is “better under this set of measures”.

- **Sensitivity to assumptions:** the statement “the recommended rule beats the current one” **depends on the tie-break rule inside the bottom-two step** (judges keep on score). Switch to the equally defensible “judges keep on fan vote”, and that advantage does not hold. **The direction (a bottom-two step raises fairness) is insensitive to the assumption; the magnitude, and “whether it dominates the current rule”, are sensitive.**

Action. Starting next season: adopt “the judges decide the bottom two” into the rules, and set the combination weights to about 1/4 judges and about 3/4 fans. **Do not simply raise the judge weight** — that trades audience engagement for fairness.

Sincerely, Team #1111111

10 References

The body of this paper relies on exactly three non-self sources: the problem statement, its data attachment, and one external dataset — the Wikimedia page-view signal tested in Section efsec:sw. Entries [1] to [3] below are those three sources; entries [4] to [6] are the AI tools and the mathematics/computing software this paper used (under the COMAP AI policy, mathematics software is also within the disclosure scope; each tool’s version, model and purpose are detailed in the “Report on Use of AI” at the end).

1. COMAP. *2026 MCM/ICM Problem C: Data With The Stars* (the Statement). The Consortium for Mathematics and Its Applications (COMAP), 2026. Referred to in the text as “the Statement”; cited for the data-description notes (notes 1a–1c, notes 5–7), the appendix giving one worked example for each of the two combination rules, and the footnote that marks the exact season of the return to the rank method as uncertain and permits assuming season 28.
2. COMAP. *2026_MCM_Problem_C_Data.csv* (data file attached to the Statement). COMAP, 2026. 53 columns $\times$ 421 rows, covering weekly judge scores and contestant identities for seasons 1–34; referred to in the text as “the Attachment”, and every reading reported in this paper was recovered from this file.
3. Wikimedia Foundation. *Wikimedia Pageviews REST API* (wikimedia.org/api/rest_v1/), retrieved 10 October 2026. Daily page views of each contestant’s English Wikipedia article, used as the external attention signal tested in Section 7; broadcast windows from the same source’s season articles and Wikidata (Q2674710). Nothing before 1 July 2015, which limits the test to seasons 21–27. Located by the AI assistant, confirmed by the team before use.

AI tools and mathematics/computing software

4. Anthropic Claude Code (v2.1.295, 2026; model: DeepSeek deepseek-flash). Used for topic selection, model selection, data cleaning, all solution code, all figures, and the drafting of the entire body text.
5. Python 3.11 (NumPy 2.3, SciPy 1.17). Mathematics/computing software; used for data cleaning, model solving and the computation of every reading (linear programming via SciPy’s HiGHS solver, rank correlations via SciPy’s spearmanr).
6. matplotlib 3.10. Plotting software; used to produce every figure in this paper.

11 Report on Use of AI

This section has no page limit and is not counted as part of the 25-page solution.

1. Anthropic Claude Code (v2.1.295, 2026; model: DeepSeek deepseek-flash, 1M context)

The agentic AI assistant used for essentially all of the intellectual work of this paper. Its use at each stage was:

- **Topic selection** – chose Problem C from the six 2027 MCM/ICM problems;
- **Model selection** – chose the inverse (feasibility / linear-programming) model, and later obtained and tested the external-proxy alternative rather than rejecting it on principle (the result is reported in Section 7);
- **Data cleaning** – read, folded and cleaned the 2026_MCM_Problem_C_Data.csv panel, and fetched the external page-view signal from the Wikimedia REST API;
- **Solution code** – wrote 100% of the ten Python scripts under code/ and fig/ (nine model / result scripts, including the two that fetch and test the external signal, and the figure script);
- **Figures** – produced 100% of the eight figures;
- **Manuscript** – drafted 100% of the body text of Sections 1–9, of the reference list and of this report.

This is a conversational LLM use (discrete prompts in, text out), so Query and Output are given below. The verbatim prompts were not kept as a standalone transcript, so they are reported as not retained rather than reconstructed.

Query: <exact wording not retained – the tasks were issued to the tool as interactive natural-language briefs within the working session> Output: <not reproduced here – the complete outputs are the manuscript sections, the eight scripts, the result files and the figures that make up this submission>

2. Python 3.11 (NumPy 2.3, SciPy 1.17) (2026)

Mathematics / computing software, used under AI-written code to clean the data and to compute every number reported in the paper. SciPy's `linprog`, with the HiGHS linear-programming solver, solved the per-contestant min/max programs that give the feasible intervals; SciPy's `spearmanr` computed the rank correlations. (No discrete query: statement form.)

3. matplotlib 3.10 (2026)

Plotting software, used under AI-written code to produce all eight figures in this paper. (No discrete query: statement form.)